Alibaba Cloud

Cross-border e-commerce for Southeast Asia

Architecture proposal

Prepared for Lumen Retail Group

Date	Revisions	Products	Catalog checked
31 October 2026	2	20	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Executive summary

Lumen Retail Group is preparing to launch a cross-border e-commerce platform targeting Southeast Asia, an initiative that demands both rapid customer acquisition and flawless technical execution. With an expected baseline of 50,000 daily active users and the potential for tenfold traffic surges during flash sales, the underlying infrastructure must be resilient enough to absorb unpredictable demand without degrading the shopping experience. A single minute of downtime or a slow-loading product page during a promotional event can permanently erode consumer trust and directly impact revenue in a highly competitive regional market. This proposal outlines an architecture designed specifically to protect your brand reputation while enabling aggressive growth across multiple countries.

The architecture we propose is built from the customer inward, starting with intelligent edge routing and robust security layers before reaching the core application and data tiers. By leveraging Alibaba Cloud's extensive network presence in Southeast Asia, we ensure that shoppers experience local-like speeds regardless of their physical location relative to your origin servers. Heavy assets such as product images are cached at the edge, dramatically reducing latency and bandwidth costs while keeping the core systems focused on transactional logic. This approach transforms geographic distance from a business risk into a managed operational detail.

To support your vision of an AI-powered shopping assistant, the design integrates advanced language model capabilities directly into the search and conversational flow. This allows customers to interact naturally with your catalog, driving higher conversion rates and larger average order values through personalized recommendations. Behind the scenes, asynchronous messaging ensures that order processing remains decoupled from the user interface, preventing checkout bottlenecks even when thousands of users transact simultaneously. The result is a modern, intelligent storefront that differentiates Lumen Retail Group from traditional e-commerce competitors.

Finally, this proposal addresses the critical need for comprehensive observability and stringent security, including the Web Application Firewall you explicitly requested. Every component is instrumented to provide real-time visibility into performance, errors, and potential threats, empowering your operations team to act before customers are impacted. Security controls are layered from the network edge down to the host level, ensuring compliance and protecting sensitive transaction data. Together, these elements form a production-ready foundation that balances cost efficiency with the high availability required for enterprise retail.

What this design delivers

Uncompromised revenue during flash sales	By automatically scaling compute resources and buffering transactions through asynchronous messaging, the platform captures every possible sale during tenfold traffic spikes. Customers will never encounter a crashed site or a timed-out checkout, ensuring that marketing investments in flash sales translate directly into realized revenue rather than lost opportunities.
Borderless customer experience	Shoppers in any Southeast Asian country will experience page load times comparable to local domestic websites. Through intelligent DNS routing, edge acceleration, and localized image caching, the architecture eliminates the friction of cross-border commerce, fostering trust and encouraging repeat purchases across diverse geographic markets.
Intelligent conversion engine	The integration of an AI shopping assistant transforms the platform from a passive catalog into an active sales associate. By understanding natural language queries and retrieving precise product information, the chatbot guides hesitant buyers toward purchase, increasing average order value and significantly reducing the burden on human customer support teams.
Fortified brand reputation	Layered security controls—from volumetric attack mitigation to application-layer bot defense—ensure that malicious actors cannot disrupt service or scrape proprietary data. This proactive defense mechanism protects Lumen Retail Group from the financial and reputational devastation associated with public-facing outages or data compromises.
Operational clarity and agility	Comprehensive observability across infrastructure, applications, and logs provides your engineering teams with immediate insight into system health. Instead of reacting to customer complaints, operations staff can identify and resolve performance anomalies proactively, accelerating development cycles and reducing the total cost of ownership.

What you told us, and how the design answers it

Each priority from our conversation, where it came from, and the part of the design that answers it.

The platform must maintain strict high availability and seamless performance during flash sales that generate up to ten times the normal traffic volume.

"10x traffic spikes during flash sales"

How the design answers it

To guarantee high availability during extreme traffic spikes, the architecture employs stateless compute instances governed by automated scaling policies, paired with cloud-native databases and in-memory caching. Traffic is distributed intelligently across multiple availability zones using Layer 7 load balancing, ensuring no single point of failure exists. Asynchronous message queuing absorbs sudden bursts of order submissions, smoothing the load on backend systems so the user experience remains uninterrupted.

Products involved

Elastic Compute Service (ECS), Auto Scaling (ESS), Application Load Balancer (ALB), PolarDB, Tair (Redis® OSS-Compatible), ApsaraMQ for RocketMQ

Shoppers across various Southeast Asian countries must experience fast page loads despite the physical distance to the central application servers.

"a cross-border e-commerce site for Southeast Asia"

How the design answers it

Cross-border latency is mitigated through a combination of authoritative DNS routing optimized for the region, dynamic edge acceleration, and localized content caching. By serving static assets from nodes physically closer to the end user and accelerating dynamic requests over optimized network paths, the architecture delivers a consistent, low-latency browsing experience. This prevents cart abandonment caused by slow load times in geographically dispersed markets.

Products involved

Alibaba Cloud DNS, Edge Security Acceleration (ESA), CDN

The site will feature extensive product imagery that must load instantly to drive conversions without overwhelming the origin infrastructure.

"lots of product images"

How the design answers it

Heavy media assets are offloaded to scalable object storage and distributed globally via a dedicated Content Delivery Network. This strategy ensures that image requests are served from the edge, drastically reducing the load on your core servers and lowering outbound bandwidth costs. Customers see high-resolution product details immediately, which is critical for maintaining engagement and trust in a visual retail environment.

Products involved

Object Storage Service (OSS), CDN

An intelligent, conversational AI shopping assistant must be integrated to guide users, answer queries, and recommend products dynamically.

"we want an AI shopping-assistant chatbot"

How the design answers it

The AI shopping assistant is powered by a managed large language model platform capable of retrieval-augmented generation over your specific product catalog. This allows the chatbot to provide accurate, context-aware responses rather than generic answers, directly influencing purchase decisions. By integrating this capability natively within the cloud environment, you avoid the complexity and latency of routing data to external third-party AI providers.

Products involved

Alibaba Cloud Model Studio, OpenSearch

The application layer requires explicit protection against web-based attacks, malicious bots, and common vulnerabilities.

"oh i also want waf"

How the design answers it

A Web Application Firewall is positioned immediately after volumetric DDoS mitigation to inspect all Layer 7 traffic. It actively blocks OWASP Top 10 threats, prevents credential stuffing, and stops automated scraping bots that target retail pricing and inventory data. This ensures that only legitimate shopper traffic reaches your compute resources, protecting both system stability and business logic.

Products involved

Web Application Firewall (WAF)

Customers require a fast, accurate, and intuitive search experience to navigate a large and growing product catalog effectively.

From your answers

How the design answers it

Given the scale of a cross-border retail operation, standard database queries are insufficient for delivering relevant search results in milliseconds. A dedicated search engine is deployed to handle complex filtering, faceting, and full-text queries, enhanced by natural language processing capabilities. This ensures users find what they are looking for quickly, directly correlating with higher conversion rates and reduced bounce rates.

Products involved

OpenSearch

Order processing workflows must be isolated from frontend traffic spikes to prevent transaction failures during peak events.

From your answers

How the design answers it

During flash sales, synchronous processing of payments, inventory updates, and fulfillment triggers would inevitably lead to timeouts and lost orders. Implementing a robust message queue decouples the checkout experience from backend fulfillment, allowing the system to accept orders rapidly and process them reliably at its own pace. This architectural pattern is essential for preserving revenue integrity when demand exceeds immediate processing capacity.

Products involved

ApsaraMQ for RocketMQ

As a cross-border retailer handling consumer transactions, the platform must adhere to strict security postures and data protection standards.

Typical for this kind of workload

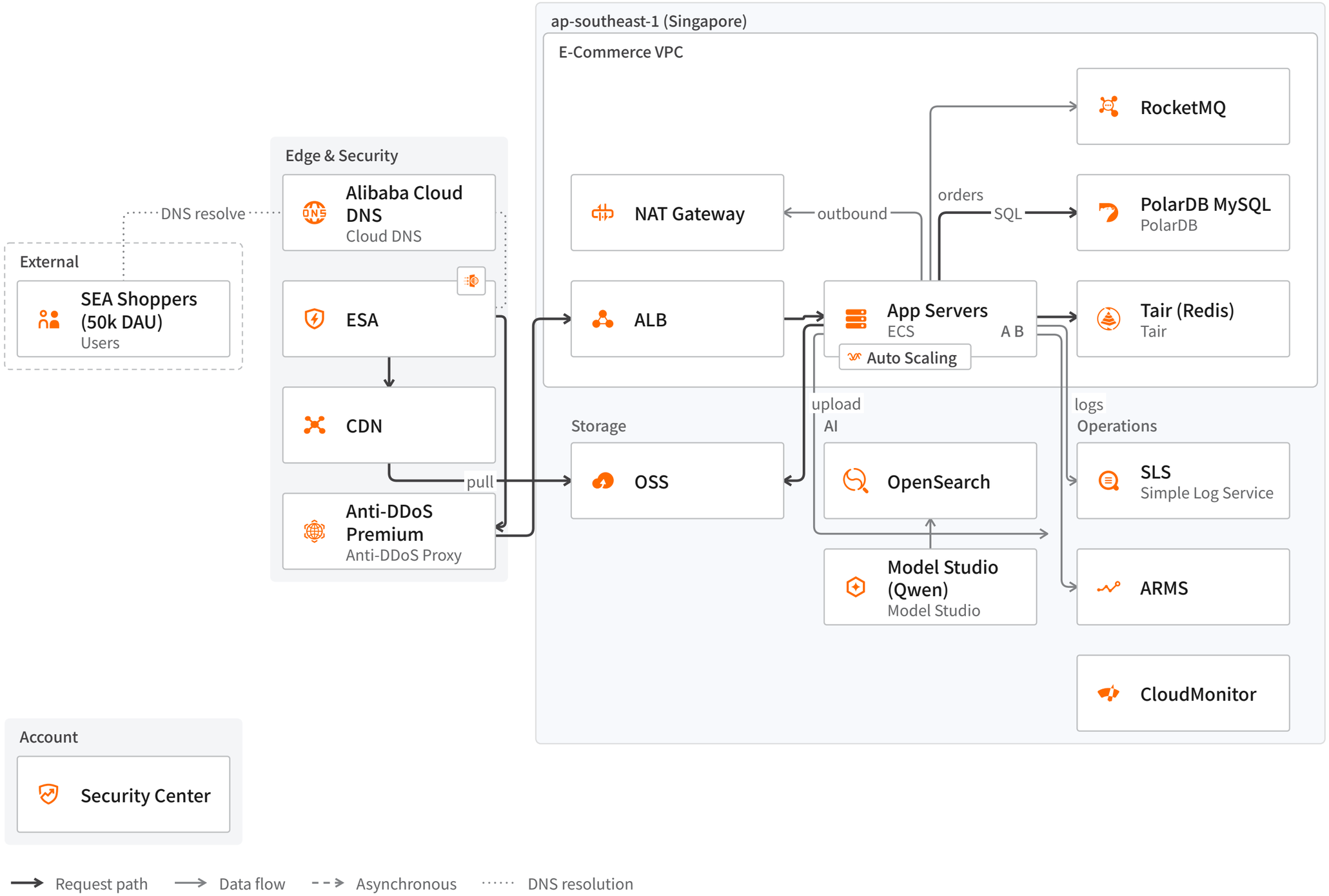
How the design answers it

Operating across multiple Southeast Asian jurisdictions requires a unified security posture that spans from the network perimeter down to individual virtual machines. Centralized threat detection, vulnerability management, and continuous compliance monitoring are implemented to satisfy regional data protection regulations. Without this holistic security framework, the business risks regulatory penalties, data breaches, and severe reputational damage.

Products involved

Security Center, Anti-DDoS Pro/Premium (Anti-DDoS Proxy)

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

SEA Shoppers (50k DAU)
External

Why it's here

End users represent the ultimate consumers of this architecture; every technical decision made in this design is measured against their expectation of speed, reliability, and security. Failing to optimize for the user experience across Southeast Asia directly results in cart abandonment and loss of market share to faster competitors.

What it does

People reaching the system over the internet from browsers or apps.

Official documentation

Documentation link not yet available

Security Center

Account · Security

Why it's here

Security Center provides continuous visibility into host-level vulnerabilities, misconfigurations, and compliance drift across the entire fleet. Ignoring host security leaves the platform exposed to internal threats and fails to meet the rigorous compliance standards expected of cross-border retailers handling payment data.

What it does

Workload protection: vulnerabilities, baselines, threat detection and Agentic SOC (SIEM/SOAR).

Official documentation

www.alibabacloud.com/product/security_center

Alibaba Cloud DNS

Alibaba Cloud DNS
Edge & Security · Domain Names and Websites

Why it's here

Alibaba Cloud DNS ensures that shoppers are routed to the most optimal entry point based on their geographic location and network conditions. Without intelligent DNS resolution, users may be directed to distant or congested endpoints, introducing unnecessary latency before the page even begins to load.

What it does

Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation

www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge & Security · Networking & CDN

Why it's here

Edge Security Acceleration optimizes the delivery of dynamic content across international borders, reducing the round-trip time for API calls and personalized page elements. Without ESA, cross-border shoppers will experience sluggish interactions during checkout and account management, leading to frustration and abandoned carts.

What it does

Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation

www.alibabacloud.com/product/esa

Web Application Firewall (WAF)

Edge & Security · On ESA

Why it's here

Web Application Firewall inspects incoming traffic for sophisticated Layer 7 threats, such as SQL injection, cross-site scripting, and automated bot scraping. Omitting WAF leaves the application logic vulnerable to exploitation, risking data breaches and unauthorized manipulation of pricing or inventory systems.

What it does

WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation

www.alibabacloud.com/product/web-application-firewall

CDN

Edge & Security · Networking & CDN

Why it's here

CDN caches heavy product images and static assets at edge nodes distributed throughout Southeast Asia. If omitted, every image request would travel back to the origin server, causing massive bandwidth costs, severe latency, and potential origin overload during traffic spikes.

What it does

Caches static content on edge nodes close to users.

Official documentation

www.alibabacloud.com/product/content-delivery-network

Anti-DDoS Pro/Premium (Anti-DDoS Proxy)

Anti-DDoS Premium
Edge & Security · Security

Why it's here
Anti-DDoS Pro/Premium shields the platform from volumetric attacks that frequently target high-profile retail flash sales. Leaving this out exposes the business to catastrophic downtime where attackers can easily overwhelm the network perimeter, halting all sales activity and damaging brand credibility.

What it does
Proxy scrubbing of large L3-L7 DDoS attacks via CNAME or IP; Premium for overseas origins.

Official documentation
www.alibabacloud.com/product/ddos

NAT Gateway

ap-southeast-1 (Singapore) · E-Commerce VPC · Networking & CDN

Why it's here
NAT Gateway allows backend compute instances to securely reach the internet for software patching and third-party API integrations without exposing them to inbound attacks. Without NAT, servers either lack necessary outbound connectivity or must be assigned public IPs, drastically expanding the attack surface of the platform.

What it does
Gives private VPC resources outbound internet (SNAT) and mapped inbound ports (DNAT).

Official documentation
www.alibabacloud.com/product/nat

Application Load Balancer (ALB)

ap-southeast-1 (Singapore) · E-Commerce VPC · Networking & CDN

Why it's here
Application Load Balancer distributes incoming web traffic evenly across multiple compute zones, ensuring no single server becomes a bottleneck. Without ALB, the architecture lacks the ability to gracefully handle node failures or distribute flash sale loads, resulting in degraded performance and downtime.

What it does
Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation
www.alibabacloud.com/product/server-load-balancer

Elastic Compute Service (ECS)

App Servers
ap-southeast-1 (Singapore) · E-Commerce VPC · Zones A, B

Why it's here
Elastic Compute Service provides the foundational, stateless compute power required to run the e-commerce application logic. Relying on rigid, non-cloud infrastructure would make it impossible to respond elastically to the tenfold traffic variations inherent in your flash sale business model.

What it does
Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation
www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

ap-southeast-1 (Singapore) · E-Commerce VPC · On App Servers

Why it's here
Auto Scaling automatically provisions and terminates ECS instances based on real-time demand, aligning infrastructure costs precisely with actual traffic. Without ESS, you would either overpay for idle capacity during normal days or suffer complete system failure during flash sales due to insufficient resources.

What it does
Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation
www.alibabacloud.com/product/auto-scaling

ApsaraMQ for RocketMQ

ap-southeast-1 (Singapore) · E-Commerce VPC · Middleware

Why it's here
ApsaraMQ for RocketMQ decouples the frontend checkout process from backend order fulfillment, acting as a shock absorber during flash sales. Without this asynchronous messaging layer, a sudden influx of orders would cascade into database locks and application crashes, directly resulting in lost revenue.

What it does
Managed RocketMQ 5.x for transactional, scheduled and ordered messaging.

Official documentation
www.alibabacloud.com/product/mq

PolarDB

PolarDB MySQL
ap-southeast-1 (Singapore) · E-Commerce VPC · Database

Why it's here

PolarDB serves as the cloud-native relational database for critical transactional data like orders and inventory, offering high availability and elastic storage. Using a traditional database would introduce replication lag and single points of failure that are unacceptable for a platform requiring strict consistency during high-speed retail events.

What it does

Cloud-native relational database (MySQL, PostgreSQL, Oracle-compatible) with compute-storage separation.

Official documentation

www.alibabacloud.com/product/polardb

Tair (Redis® OSS-Compatible)

ap-southeast-1 (Singapore) · E-Commerce VPC · Database

Why it's here

Tair provides ultra-fast, in-memory caching for shopping carts, user sessions, and flash sale stock counters. Bypassing this cache and hitting the primary database for every read operation during a flash sale would cause immediate database exhaustion and widespread checkout failures.

What it does

Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation

www.alibabacloud.com/product/tair

Object Storage Service (OSS)

ap-southeast-1 (Singapore) · regional services · Storage

Why it's here

Object Storage Service offers infinitely scalable, highly durable storage for the vast library of product images required for the catalog. Storing these assets on compute instances or traditional block storage would be prohibitively expensive, difficult to manage, and impossible to integrate efficiently with the CDN.

What it does

Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation

www.alibabacloud.com/product/object-storage-service

OpenSearch

ap-southeast-1 (Singapore) · regional services · AI & Machine Learning

Why it's here

OpenSearch delivers sub-second, highly relevant product search results and powers the retrieval mechanism for the AI assistant. Relying on basic database text matching would yield poor search relevance, frustrating users and severely limiting the effectiveness of the AI shopping chatbot.

What it does

Managed search, including vector search and LLM-based conversational search (RAG) editions.

Official documentation

www.alibabacloud.com/product/opensearch

Alibaba Cloud Model Studio

Model Studio (Qwen)

ap-southeast-1 (Singapore) · regional services · AI & Machine Learning

Why it's here

Alibaba Cloud Model Studio hosts and manages the large language models necessary to power the AI shopping assistant. Attempting to build and host these models independently would require prohibitive capital expenditure and specialized talent, delaying your time-to-market significantly.

What it does

Managed model APIs for Qwen (such as qwen3.8-max) and third-party models; OpenAI-compatible.

Price

Pay per token; qwen3.8-max $2 per 1M input and $6 per 1M output tokens (Singapore); 1M free tokens per model for 90 days, as of 2026-10-31

Official documentation

www.alibabacloud.com/product/modelstudio

Simple Log Service (SLS)

ap-southeast-1 (Singapore) · regional services · Storage

Why it's here

Simple Log Service centralizes all application and infrastructure logs for rapid troubleshooting and audit compliance. Operating without centralized logging means that when an issue occurs during a critical sale, engineers will waste valuable time searching disparate systems instead of fixing the problem.

What it does

Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation

www.alibabacloud.com/product/log-service

Application Real-Time Monitoring Service (ARMS)

ap-southeast-1 (Singapore) · regional services · Middleware

Why it's here

Application Real-Time Monitoring Service provides deep tracing into microservices, pinpointing exactly where latency or errors occur in the transaction flow. Without APM, performance degradation becomes a black box, making it impossible to guarantee the fast user experience required for retail conversion.

What it does

APM, browser and synthetic monitoring, tracing and alerting for applications.

Official documentation

www.alibabacloud.com/product/arms

CloudMonitor

ap-southeast-1 (Singapore) · regional services · Migration & O&M Management

Why it's here

CloudMonitor tracks infrastructure metrics and triggers the Auto Scaling policies that keep the site alive during traffic spikes. Missing this component means the auto-scaling engine operates blind, failing to add capacity in time to prevent flash sale outages.

What it does

Resource metrics, alarms and site monitoring; CMS 2.0 adds APM and AI observability.

Official documentation

www.alibabacloud.com/product/cloud-monitor

Considerations and next steps

Trade-offs and how they are handled

Deploying both Edge Security Acceleration and a traditional CDN introduces overlapping capabilities that may increase initial configuration complexity and monthly operational costs. However, relying solely on one may leave gaps in either dynamic acceleration or static caching efficiency.

How it is handled

We will configure ESA strictly for dynamic API and personalized content routing, while reserving the CDN exclusively for static product images and assets. This clear separation of concerns optimizes cost and simplifies ongoing management.

Utilizing Tair for in-memory caching alongside PolarDB for persistent storage creates a dual-write scenario where data consistency must be carefully managed, especially for flash sale inventory counts where overselling is a direct revenue risk.

How it is handled

The architecture implements atomic decrement operations within Tair for flash sale stock, treating the cache as the source of truth for high-speed inventory gating, while asynchronously reconciling with PolarDB to ensure eventual consistency without blocking the user.

Integrating Alibaba Cloud Model Studio for the AI chatbot adds variable inference costs based on user interaction volume, which can become unpredictable during viral flash sale events compared to fixed-cost traditional infrastructure.

How it is handled

We will implement token usage quotas, rate limiting per user session, and fallback mechanisms to standard keyword search if AI costs exceed predefined thresholds, ensuring predictable budgeting while maintaining the premium experience.

Decoupling order processing with ApsaraMQ for RocketMQ improves resilience but introduces eventual consistency, meaning customers might not see immediate order confirmation on their screen the millisecond they click buy.

How it is handled

The frontend will display an immediate 'Order Received' acknowledgment upon successful message publication to the queue, while background workers process the actual fulfillment, providing a seamless user perception without sacrificing backend stability.

Implementing comprehensive observability through SLS, ARMS, and CloudMonitor generates significant telemetry data volumes, which incurs storage and ingestion costs that scale with the 10x traffic spikes during promotional events.

How it is handled

Log retention policies will be tiered, keeping high-fidelity trace data for a short duration (e.g., 7 days) for immediate debugging, while aggregating and archiving long-term metrics to lower-cost storage for compliance and trend analysis.

Next steps

1. Schedule a technical deep-dive workshop with your engineering leads to review the API integrations required for the OpenSearch and Model Studio components.
2. Conduct a simulated flash sale load test using Auto Scaling and RocketMQ to validate the 10x traffic handling capacity and measure exact cost projections.
3. Finalize the Web Application Firewall rule sets and Anti-DDoS thresholds based on your historical traffic patterns and specific regional threat intelligence.
4. Review data residency and compliance requirements for each target Southeast Asian country to ensure Security Center and log retention policies meet local regulations.
5. Establish a phased rollout plan, beginning with a beta launch in a single market to tune CDN caching rules and ESA dynamic routing before regional expansion.
6. Define operational runbooks and alerting thresholds within CloudMonitor and ARMS to empower your support team to respond autonomously to infrastructure events.

Design record

What the design rests on: the requirements pinned in the conversation, constraints, assumptions and open questions, with the notes from the first drawing.

The design

Let's architect your cross-border e-commerce platform for Southeast Asia, building from users inward. First, the edge layer to handle global acceleration, DDoS protection, and caching product images close to users. Now the core region with a VPC, highly available load balancers, and auto-scaling compute for 10x flash sales. For data, we need cloud-native relational DB, Redis cache for flash sales, object storage for images, and search. Adding the AI shopping assistant chatbot using Model Studio for Qwen LLM and RocketMQ for async order events. Finally, observability, security, and network egress to complete the highly available architecture.

20 products, 17 connections, 2 revisions

Pinned requirements

No requirements were pinned.

Constraints

No region or product constraints were set.

Assumptions

No assumptions were recorded.

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
SEA Shoppers (50k DAU)	Alibaba Cloud DNS	DNS resolve	DNS resolution
Alibaba Cloud DNS	ESA		DNS resolution
ESA	Anti-DDoS Premium		Data
ESA	CDN	images	Data
Anti-DDoS Premium	ALB	HTTPS	Data
ALB	App Servers		Data
App Servers	PolarDB MySQL	SQL	Data
App Servers	Tair (Redis)	cache	Data
App Servers	OSS	upload	Data
CDN	OSS	pull	Data
App Servers	OpenSearch	query	Data
App Servers	Model Studio (Qwen)	chat API	Data
Model Studio (Qwen)	OpenSearch	RAG	Data
App Servers	RocketMQ	orders	Data
App Servers	SLS	logs	Data
App Servers	ARMS	traces	Data
App Servers	NAT Gateway	outbound	Data

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 35.8 s

Ask

We are launching a cross-border e-commerce site for Southeast Asia: about 50k daily active users, 10x traffic spikes during flash sales, product search, lots of product images, and we want an AI shopping-assistant chatbot. It must be highly available.

Changes

+ Users · SEA Shoppers (50k DAU) · in External
+ Cloud DNS · Alibaba Cloud DNS · in Edge & Security
+ Anti-DDoS Proxy · Anti-DDoS Premium · in Edge & Security
+ WAF · on ESA
+ ESA · in Edge & Security
+ CDN · in Edge & Security
+ 20 connections
+ ALB · in E-Commerce VPC
+ ECS · App Servers (AZ-A)
+ Auto Scaling · on ECS
+ PolarDB · PolarDB MySQL · in E-Commerce VPC
+ Tair (Redis) · in E-Commerce VPC
+ OSS · in E-Commerce VPC
+ OpenSearch · in E-Commerce VPC
+ Model Studio · Model Studio (Qwen) · in E-Commerce VPC
+ RocketMQ · in E-Commerce VPC
+ SLS · in E-Commerce VPC
+ ARMS · in E-Commerce VPC
+ CloudMonitor · in E-Commerce VPC
+ Security Center · in E-Commerce VPC
+ NAT Gateway · in E-Commerce VPC

Revision 2

Drawn in 3.5 s

Ask

oh i also want waf

Changes

No change to the diagram.